

A24

IN THE *FILM* INDUSTRY

UNDERSTANDING HOW A PRESTIGE INDIE STUDIO COMPARES TO
THE BROADER FILM INDUSTRY THROUGH DATA

BANA 212: Data & Programming
for Analytics

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A24

- Founded in 2012 as an independent film and television studio
- Known for distinctive storytelling, creative risk-taking & critical acclaim
- Released notable titles such as Hereditary & Everything Everywhere All at Once
- Recognized for achieving strong commercial outcomes despite smaller budgets

“Hollywood tends to hide complex ideas.
A24 embraces them”

– ALEX GARLAND (DIRECTOR OF EX MACHINA)

WRITTEN AND DIRECTED BY
ALEX GARLAND THE WRITER OF
28 DAYS LATER

EX_MACHINA

THERE IS NOTHING MORE HUMAN THAN THE WILL TO SURVIVE

FILM4 COMING SOON

DOLBY
DIGITAL

AMERICAN
FILM INSTITUTE

Adam Sandler

A SAFDIE BROTHERS FILM

Uncut Gems

DECEMBER



WHY THIS MATTERS

Hollywood

- Large budgets
- Extensive financial forecasting
- Proven formulas
- Higher commercial predictability

A24

- Lean budgets
- Smaller production scale
- Prioritizes originality
- Lower commercial predictability

- Film production depends heavily on box office success and is high risk.
- Identifying key predictors of revenue can improve investment decisions, genre positioning, and overall film production or marketing strategies.

QUESTION:

Do A24 films meaningfully differ from the broader film industry in ways that influence box office performance?

DATA OVERVIEW

- IMDb_movies (Main)
 - ◉ Kaggle dataset containing 10,178 films scraped from IMDB.com
 - ◉ Includes title, release date, genre, cast, rating, budget, revenue and country
 - ◉ Used for main analysis
- A24_flag
 - ◉ Dataset obtained from A24 studio fans containing 182 films by A24
 - ◉ Includes movie titles and release year
 - ◉ Used to flag A24 movies in main dataset





LET THE

FESTIVITIES

BEGIN

JULY 3

from ARI ASTER
director of HEREDITARY

MIDSOMMAR

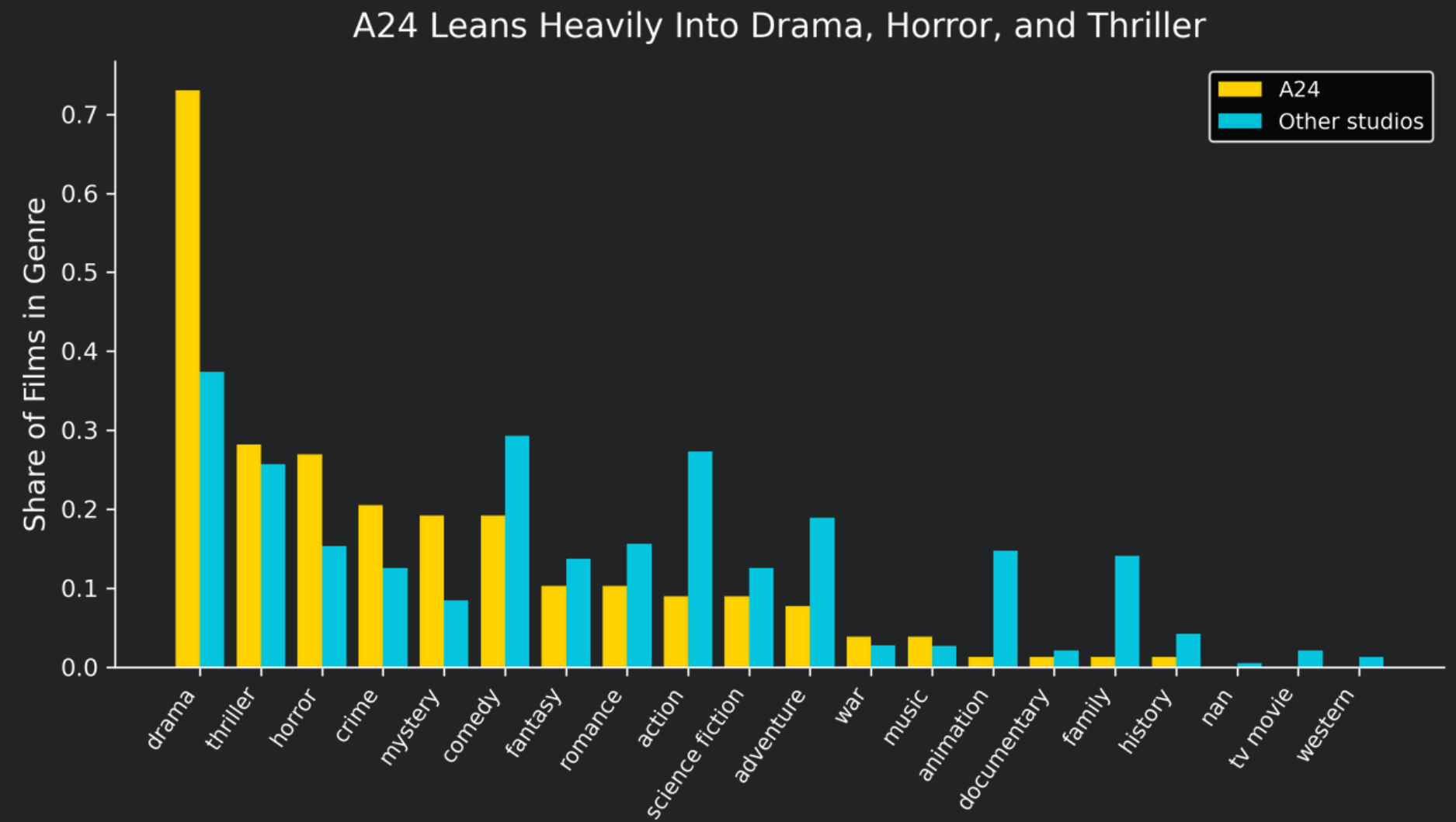
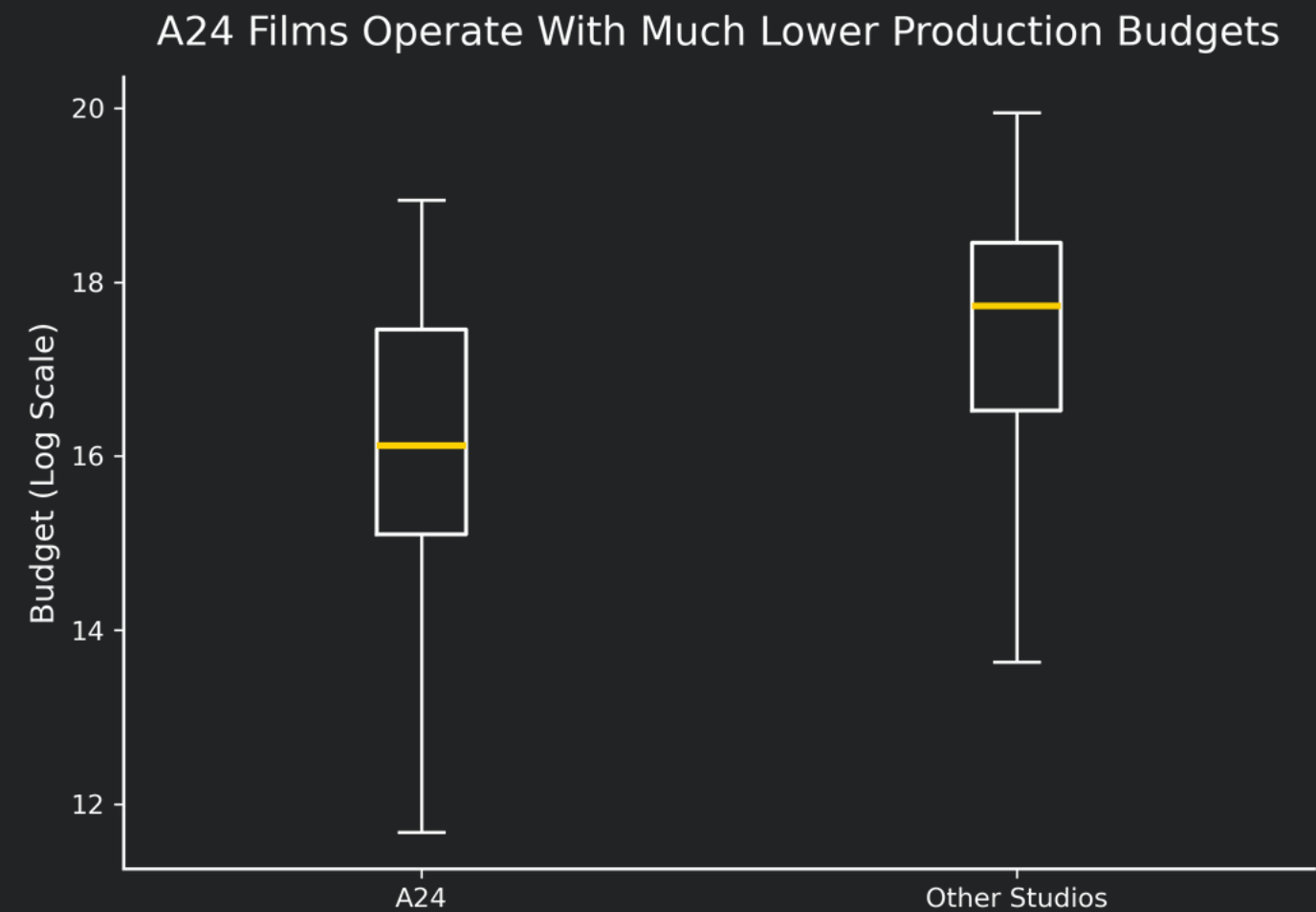
A24

THIS FILM IS NOT YET RATED

DATA PREPROCESSING

- Merged IMDb dataset with A24 film list
 - ◉ Matched on title & year
- Removed rows with missing or zero budget or revenue
- Extracted release_year and standardized titles
- Normalized IMDb score to 0–100 scale
- One-hot encoded genres
 - ◉ Cleaned duplicates
- Created is_A24 dummy variable
 - ◉ =1 if it is a A24 production, =0 if otherwise
- Log-transformed budget and revenue to reduce skew

EXPLORATORY DATA ANALYSIS

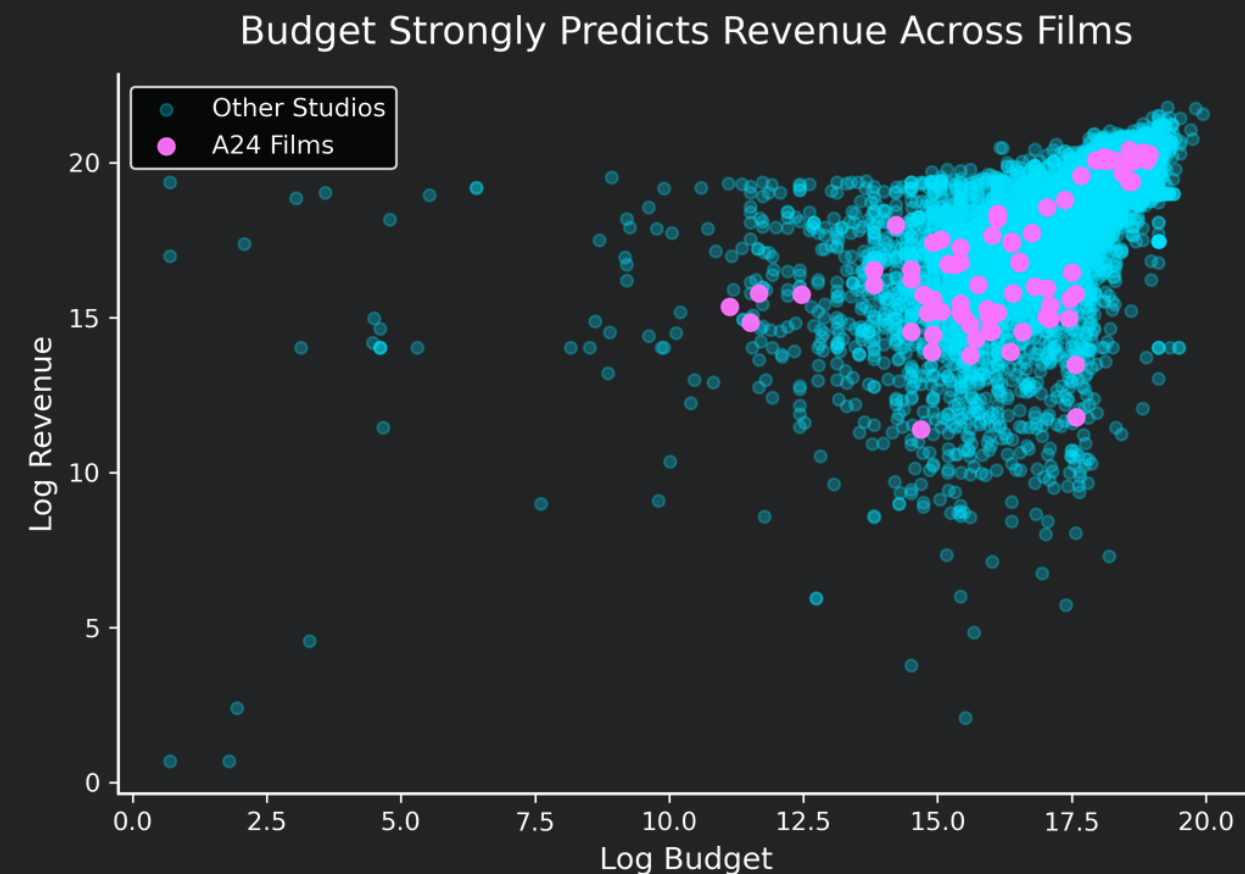


- A24 operates with much smaller budgets
- Strong positive relationship between budget & revenue
- Genre profile: Drama, Horror, Thriller dominate

LINEAR REGRESSION

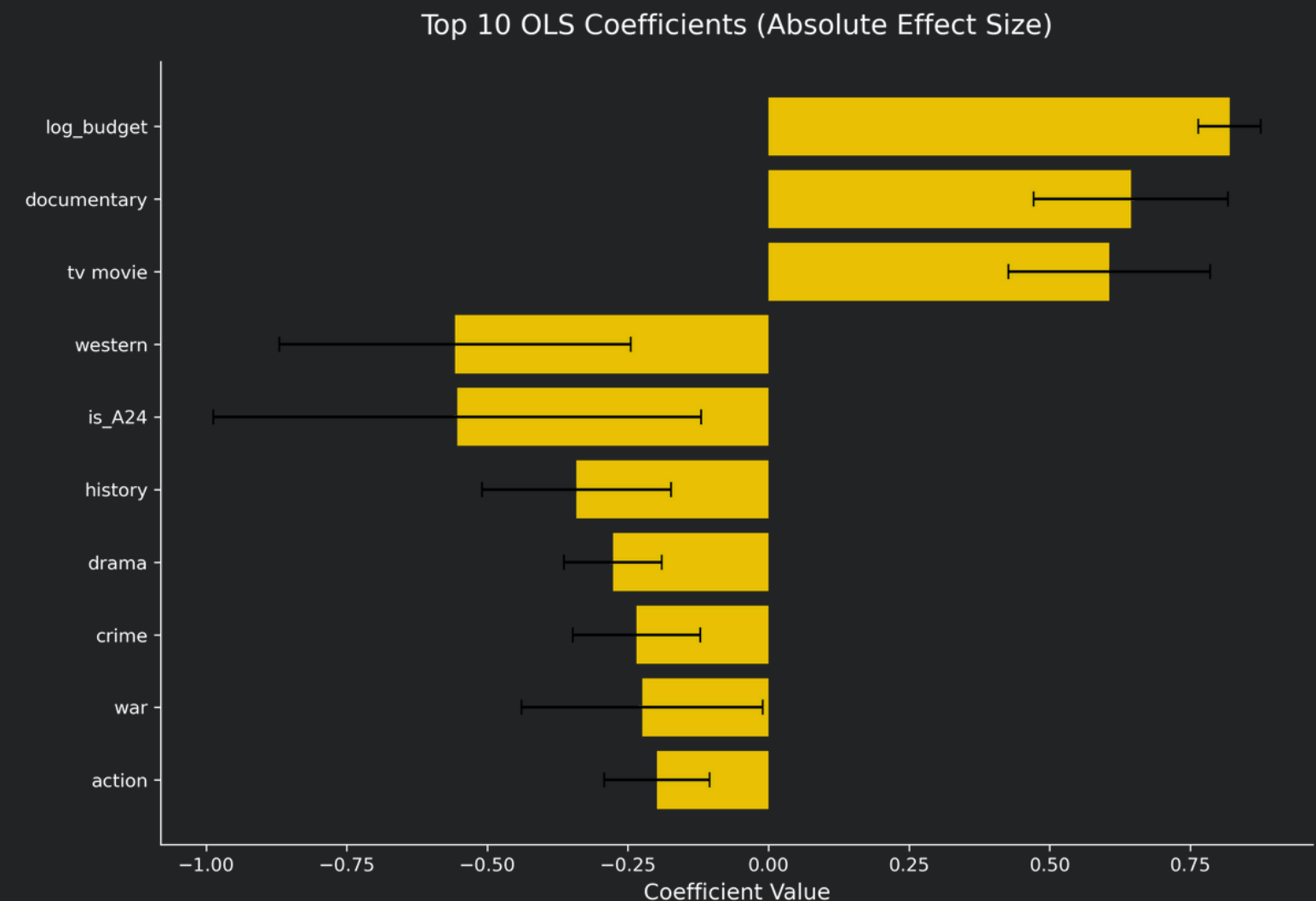
MODEL SETUP

- **Method:** Ordinary Least Square (OLS) Regression
- **Target variable:** log_revenue
- **Predictors:**
 - log(budget)
 - release_year
 - IMDb score
 - is_A24
 - genre dummies
- **Purpose:** Provide a simple, interpretable baseline for how pre-release features relate to revenue



KEY FINDINGS

- $R^2 \approx 0.42 \rightarrow$ the model explains ~42% of variation in log(revenue)
- RMSE ≈ 1.70 in log space $\rightarrow$ moderate predictive accuracy
- log_budget has the **strongest** and **most significant positive** effect
- IMDb score and release_year show moderate, significant contributions
- is_A24 has a small, non-dominant effect once budget, ratings, and genre are controlled



RANDOM FOREST REGRESSION

MODEL SETUP

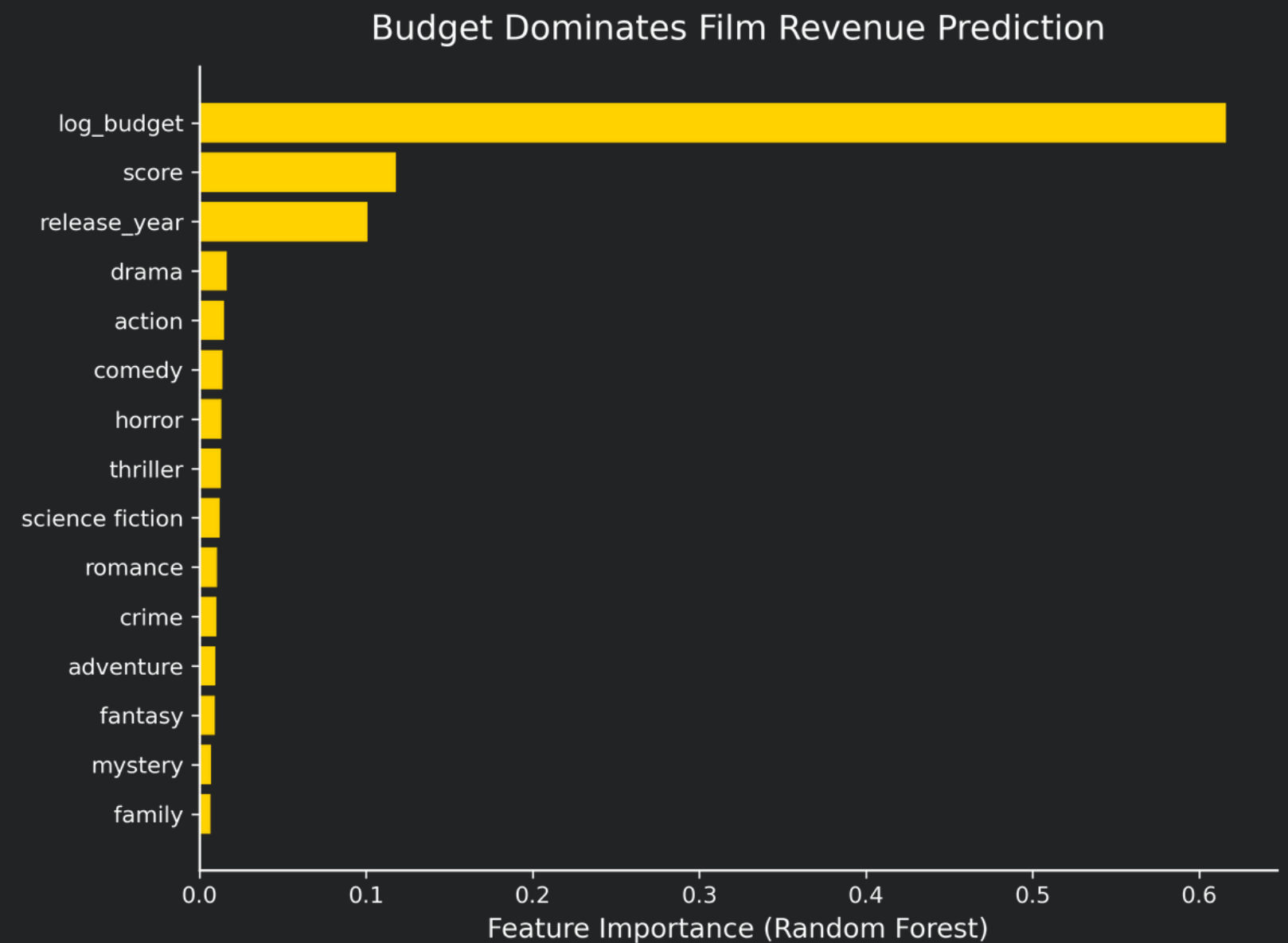
- **Purpose:** Captures nonlinear relationships & interactions that OLS cannot
- **Approach:** Ensemble of decision trees

EVALUATION

- $R^2 \approx 0.55 \rightarrow$ better than OLS
- Lower RMSE than the linear model
- 10-fold CV $R^2 \approx 0.53 \pm 0.07 \rightarrow$ stable across folds

INTERPRETATION

- Confirms budget as the dominant driver
- Ratings and release_year add incremental predictive value
- Genre and A24 identity have relatively small marginal effects



MODEL PERFORMANCE SUMMARY

PERFORMANCE OVERVIEW

- **Linear Regression**
 - $R^2 \approx 0.42$
 - $RMSE \approx 1.69$
- **Random Forest**
 - Test $R^2 \approx 0.55$
 - $RMSE \approx 1.46$ (log space)
 - 10-fold CV $R^2 \approx 0.53 \pm 0.07$

ROBUSTNESS CHECKS

- Multiple train/test splits → RF wins consistently
- Feature importance stable across runs
- Model results not dependent on one lucky split

KEY TAKEAWAYS

- Multiple train/test splits → RF wins consistently
- Findings on budget, ratings, and A24's role hold across models
- Budget remains the primary driver

THE BRUTALIST



Mia Goth

INSIGHTS

#1. BUDGET = STRONGEST DRIVER

- Largest effect across all models
- Consistent in feature importance

#2. RATINGS & RELEASE TIMING MATTER

- Higher critic/IMDb scores → better performance
- Recent release years perform slightly better

#3. GENRE EFFECTS ARE MODEST

- Higher-impact genres: Drama, Action, Comedy, Horror, Thriller
- Lower-impact genres: Family, Animation, History, Western
- Genre alone does not strongly determine revenue.

a film by Ti West
"Pearl"
*An Xtraordinary
Origin Story*

A24 PRESENTS A LITTLE LAMB PRODUCTION A FILM BY TI WEST "PEARL" MIA GOTH DAVID CORENSWET
TANDI WRIGHT MATTHEW SUNDERLAND EMMA JENKINS-PURRO CASTING BY STU TURNER KARLA SPIKA
MUSIC BY TYLER BATES AND TIM WILLIAMS COSTUME DESIGNER MALGOSIA TURZANSKA EDITED BY TI WEST
PRODUCTION DESIGNER TOM HAMMOCK DIRECTOR OF PHOTOGRAPHY ELIOT ROCKETT LINE PRODUCER JARED CONNOR
EXECUTIVE PRODUCERS MIA GOTH PETER PHOK SAM LEVINSON ASHLEY LEVINSON SCOTT MESCUDI
DENNIS CUMMINGS KARINA MANASHIL PRODUCED BY JACOB JAFFKE, p.g.a. TI WEST, p.g.a.
KEVIN TUREN HARRISON KREISS BASED ON CHARACTERS CREATED BY TI WEST
WRITTEN BY TI WEST & MIA GOTH DIRECTED BY TI WEST

A24

IMPLICATIONS

✦ PRODUCTION STUDIOS

- Budget strategy matters → modest increases can boost revenue
- Prioritize projects with strong ratings potential
- Genre less critical than financial positioning

✦ STREAMERS & DISTRIBUTORS

- A24-style films offer strong value per dollar
- High-rating, low-budget films = efficient returns
- Good targets for acquisition & recommendation algorithms

✦ INVESTORS

- Prestige & independent films can perform competitively
- Returns improve when budgets are controlled
- Lower-budget, high-rating films present attractive risk–reward balance



HEREDITARY

*Every family tree
hides a secret.*



LIMITATIONS

#1. DATA GAPS

- Missing studio labels for non-A24 films
- IMDb dataset incomplete → not all global releases included
- Revenue data excludes streaming, TV, awards boosts

#2. MISSING PREDICTORS

- No marketing spend, star power, or theater count
- Limited rating sources (critics vs. audience breakdown not complete)
- Review scores vary across platforms; not standardized

#3. MODEL CONSTRAINTS

- A24 films make up <1% of dataset → model driven by industry-wide patterns
- Insights are comparative, not predictive for A24 specifically

BODIES
BODIES
BODIES

RECOMMENDATIONS

for PRODUCTION STUDIOS

- Use budget modeling to guide greenlight decisions
- Track genre, budget, and score indicators early in development

for STREAMING PLATFORMS & DISTRIBUTORS

- Prioritize high-rating, low-budget films for acquisition
- Focus on titles with strong efficiency (revenue relative to budget)

for ANALYSTS & FUTURE STUDIES

- Combine IMDb with studio metadata + marketing data
- Add predictors: star power, theater count, detailed review sources

A24 presents

*Dakota Johnson
Chris Evans and
Pedro Pascal*
in

Find your
perfect match.



Materialists

From academy award® nominee
Celine Song

June 13, 2025

A24 PRESENTS A ZAM PRODUCTION A KILLER FILMS PRODUCTION AN ACCESS ENTERTAINMENT AND IFC FILM A FILM BY CELINE SONG "MATERIALISTS" DAKOTA JOHNSON CHRIS EVANS AND PEDRO PASCAL COSTUME DESIGNER DOUGLAS AUBEL
EXECUTIVE PRODUCERS DANIEL PERMOUEHTON PRODUCED BY KATINA DANABASSIS PRODUCED BY KEITH FRAASE WRITTEN BY ANTHONY CASPARRO DIRECTED BY SHADIEH KIRCHNER CASTING BY TAYLOR SHUNG LEN BLAVATNIK EDITOR DANNY COHEN EXECUTIVE PRODUCERS ANDREA SCARSO PRODUCED BY DAVID HINDJOSA CHRISTINE VACHON PAMELA KOFFLER CELINE SONG PRODUCED BY CELINE SONG

THANK YOU